

Sprint Action Plan: Person 3: Retrieval Lead

Passkey Suite, Needle-in-a-Haystack & Statistical Non-Inferiority • Adaptive Attention Computation Sprint

Assigned Role: Person 3: Retrieval Lead	Estimated Workload: 9.0 Hours	Sprint Timeline: 0.0h – 9.5h
Core Discipline: Attention Quality Evaluation / Synthetic Probing	Primary Handoff To: Person 6 (Synthesis Lead)	Target Hardware: Dual T4 (16GB)

1. Role Mission & Strategic Objective

Evaluate long-range information retrieval and key-value binding fidelity using synthetic distractor suites. Quantify the retrieval accuracy collapse of linear attention and execute pre-registered bootstrap non-inferiority tests.

2. Input Dependencies & Final Deliverables

Type	Item / Specification	Source / Destination	Deadline / Gate
INPUT	Frozen CSV schema contract	Person 6 (Synthesis Lead)	Hour 1.0
INPUT	Linear and FAVOR+ attention implementations	Person 1 (Theory Lead)	Hour 1.0
OUTPUT	Retrieval benchmark notebook (02_Passkey_Suite.ipynb)	Team Repository	Hour 7.5
OUTPUT	Retrieval accuracy CSVs (retrieval_raw.csv)	Person 6 (Synthesis Lead)	Hour 7.5
OUTPUT	Bootstrap non-inferiority audit table & heatmaps	Person 6 (Synthesis Lead)	Hour 8.0

3. Technical Specifications & Contract Signature

Key Code / Mathematical Contract for Person 3: Retrieval Lead:
• Class Coding: 16 balanced orthonormal classes via QR decomposition (DV=64, C=16); chance level = 6.25% • Distractor Protocol: Query is target key with noise (noise=0.3); all other N-1 keys act as distractors • Non-Inferiority Margin: $\delta = 0.05$; Paired difference $\Delta = \text{acc_linear} - \text{acc_softmax}$ across shared random seeds • Resampling: 2,000 bootstrap iterations over independent retrieval tasks to construct 95% Confidence Intervals

4. Hour-by-Hour Master Action Checklist — Person 3: Retrieval Lead

Window	Specific Actions & Step-by-Step Instructions	Deliverable / Evidence	Status
0.0 – 1.0 h	Implement synthetic Passkey token generator with fixed-dimensional orthonormal class codes.	Passkey data generation harness	[] Done
1.0 – 3.0 h	Run baseline Softmax retrieval grid across N = 64 to 4,096; verify 100% accuracy sanity check.	Softmax baseline validation (100% acc)	[] Done
3.0 – 5.5 h	Execute Linear Attention (ReLU+1) Passkey sweep; document collapse from 57.7% (N=64) to 14.2% (N=1024).	Linear attention retrieval collapse curve	[] Done
5.5 – 7.5 h	Benchmark FAVOR+ sweeps across $r \in \{64, 256, 1024\}$ and key norm scales (2 to 24); test local vs global targets.	FAVOR+ rescue attempt data table	[] Done
7.5 – 8.5 h	Run 2,000 bootstrap iterations; compute 95% CI for Δ ; export noninferiority.csv and 2D heatmaps to Person 6.	noninferiority.csv + Fig 2 & Fig 3 charts	[] Done
8.5 – 9.5 h	Audit retrieval figures and non-inferiority intervals in presentation deck; participate in dry-run.	Presentation audit sign-off	[] Done

5. Operational Guardrails & Failure Protocols

Constraint / Risk	Enforcement Rule & Exact Action
One-Hot Dimension Inflation	Never use N-dimensional one-hot values; increasing d_v with N invalidates scaling comparisons.
Weak Baseline logit scale	Verify softmax achieves $\geq 95\%$ accuracy on baseline tasks before interpreting relative drops.
Permutation Equivariance	Control key distractor counts separately from target token sequence position.

6. Definition of Done (DoD) & Sign-Off Checklist

[]	Baseline sanity check verified: Softmax achieves 100.0% accuracy on retrieval tasks.
[]	Linear collapse accurately documented (14.2% at N=1024, 9.5% at N=4096).
[]	Bootstrap 95% CI calculated, confirming rejection of non-inferiority ($\Delta = -0.86$ to -0.91).
[]	retrieval_raw.csv and noninferiority.csv delivered to Person 6.